

Problem Statement - 3

Title of Problem Statement: Design & Development of a High-Sensitivity Microbarometer Infrasonic sensor

Background: The Infrasonic sensors are precision instruments designed to detect and measure low frequency atmospheric pressure waves, known as infrasonic, that fall below the range of human hearing, typically under 20 Hz. These waves can travel long distances through the atmosphere and are produced by a variety of natural and human-made sources including distant Industrial explosions, volcanic eruptions, severe weather systems, meteors, rocket launches, and other energetic phenomena. Detection and analysis of these signals are important for atmospheric monitoring, geophysical research, disaster warning systems and security applications.

Description: It is required to design and develop a high-sensitivity atmospheric microbarometer Infrasonic sensor capable of measuring infrasonic pressure fluctuations in the frequency range of approximately 0.01 Hz to 20 Hz.

The sensor should address the complete hardware architecture, including:

- (a). Pressure sensing mechanism.
- (b). Mechanical transducer design.
- (c). Differential pressure measurement technique.
- (d). Low-noise analog front-end electronics.
- (e). Temperature compensation.
- (f). Long-period pressure equalization system.
- (g). Environmental enclosure.
- (h). Wind-noise reduction interface.
- (i). Calibration methodology.

The design should aim to detect very small pressure variations while maintaining long-term stability, low drift, and high signal fidelity. The data acquisition (digitizer) and real time waveform display & analysis software available in open market to be included to demonstrate complete functional sensor system.

Expected Solution/Deliverables: The prototype infrasonic sensor should have high sensitivity, long-term stability and low-noise signal condition to measure infrasonic signals accurately. Sensor should demonstrate:

- (a). Detection of low-frequency pressure signals.

- (b). Laboratory characterization of frequency response.
- (c). Noise floor measurements.
- (d). Sensitivity estimation.
- (e). Stability testing.

The evaluation will be conducted based on the achievement of the following parameters

S.No.	Parameter	Target
(a).	Frequency Band	0.01–20 Hz
(b).	Pressure Sensitivity	20 mV/Pa
(c).	Pressure Resolution	Better than 0.05 Pa
(d).	Output Voltage range	Max 24 V peak to Peak
(e).	Dynamic Range	>100 dB
(f).	Sampling Rate	≥50 Hz
(g).	Compatibility	With 24 bit digitizer

The digitizer and data acquisition software (available in open market) for real time waveform display & analysis will be arranged by candidates themselves to demonstrate the complete functional sensor system.

Organisation: National Technical Research Organisation

Category: Hardware

Theme: Smart Automation

Youtube Link/Video Link: Nil

Dataset Link: Nil